Year 10 Foundation

REVISION PACK

Number 1

Name: _____

Class: _____

Number 1 · Working with fractions (Lesson 1.1)

Worked example: adding fractions

Work out $\frac{2}{3} + \frac{1}{4}$

Step 1	Make the bottoms the same	$\frac{8}{12} + \frac{3}{12}$
Step 2	Add the tops	$\frac{11}{12}$
Step 3	Answer	$\frac{11}{12}$

Practice: adding fractions

1.1 Work out $\frac{3}{5} + \frac{1}{4}$

1.2 Work out $\frac{5}{6} + \frac{1}{8}$

1.3 Work out $\frac{7}{10} + \frac{1}{6}$

1.4 Work out $\frac{2}{7} + \frac{1}{3}$

1.5 Work out $\frac{1}{2} + \frac{2}{9}$

1.6 Work out $\frac{3}{8} + \frac{1}{3}$

Number 1 · Working with fractions (Lesson 1.1)

Worked example: multiplying fractions

Work out $\frac{3}{4} \times \frac{8}{9}$

Give your answer in its simplest form.

Step 1	Multiply the tops	$3 \times 8 = 24$
Step 2	Multiply the bottoms	$4 \times 9 = 36$
Step 3	Divide top and bottom by 12	$\frac{24}{36} = \frac{2}{3}$
Step 4	Answer	$\frac{2}{3}$

Practice: multiplying fractions

1.7 Work out $\frac{5}{6} \times \frac{3}{10}$. Give your answer in its simplest form.

1.8 Work out $\frac{4}{7} \times \frac{7}{12}$. Give your answer in its simplest form.

1.9 Work out $\frac{2}{5} \times \frac{15}{16}$. Give your answer in its simplest form.

1.10 Work out $\frac{3}{4} \times \frac{2}{9}$. Give your answer in its simplest form.

1.11 Work out $\frac{5}{8} \times \frac{4}{15}$. Give your answer in its simplest form.

1.12 Work out $\frac{7}{9} \times \frac{3}{14}$. Give your answer in its simplest form.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: dividing fractions

Work out $\frac{5}{6} \div \frac{2}{3}$

Give your answer as a mixed number.

Step 1 Flip the 2nd, then multiply	$\frac{5}{6} \times \frac{3}{2} = \frac{15}{12}$
Step 2 Divide top and bottom by 3	$\frac{5}{4}$
Step 3 Answer	$1\frac{1}{4}$

Practice: dividing fractions

1.13 Work out $\frac{3}{4} \div \frac{2}{5}$. Give your answer as a mixed number.

1.14 Work out $\frac{7}{8} \div \frac{1}{3}$. Give your answer as a mixed number.

1.15 Work out $\frac{9}{10} \div \frac{3}{5}$. Give your answer as a mixed number.

1.16 Work out $\frac{5}{6} \div \frac{4}{9}$. Give your answer as a mixed number.

1.17 Work out $\frac{2}{3} \div \frac{4}{7}$. Give your answer as a mixed number.

1.18 Work out $\frac{8}{9} \div \frac{2}{3}$. Give your answer as a mixed number.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: fractions of amounts

Work out $\frac{2}{5}$ of 35

Step 1	Divide by the bottom	$35 \div 5 = 7$
Step 2	Multiply by the top	$7 \times 2 = 14$
Step 3	Answer	14

Practice: fractions of amounts

1.19 Work out $\frac{5}{6}$ of 42

1.20 Work out $\frac{2}{9}$ of 81

1.21 Work out $\frac{4}{5}$ of 75

1.22 Work out $\frac{3}{7}$ of 56

1.23 Work out $\frac{5}{8}$ of 96

1.24 Work out $\frac{2}{3}$ of 57

Number 1 · Working with fractions (Lesson 1.1)

Worked example: fractions, decimals and percentages

Write $\frac{3}{8}$

(a) as a decimal. (b) as a percentage.

(a)	**Step 1** Top ÷ bottom	$3 \div 8 = 0.375$
(b)	**Step 1** × 100 for a percentage	$0.375 \times 100 = 37.5\%$

Practice: fractions, decimals and percentages

1.25 Write $\frac{2}{5}$

(a) as a decimal. (b) as a percentage.

1.26 Write $\frac{7}{20}$

(a) as a decimal. (b) as a percentage.

1.27 Write $\frac{5}{8}$

(a) as a decimal. (b) as a percentage.

1.28 Write $\frac{3}{4}$

(a) as a decimal. (b) as a percentage.

1.29 Write $\frac{9}{25}$

(a) as a decimal. (b) as a percentage.

1.30 Write $\frac{1}{8}$

(a) as a decimal. (b) as a percentage.

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

Worked example: order of operations

Work out

(a) $9 + 2 \times 6$ (b) $(10 - 4)^2 \div 3$

(a)	**Step 1**	Multiply first	$2 \times 6 = 12$
	Step 2	Then add	$9 + 12 = 21$
(b)	**Step 1**	Brackets first	$10 - 4 = 6$
	Step 2	Then the power	$6^2 = 36$
	Step 3	Then divide	$36 \div 3 = 12$

Practice: order of operations

2.1 Work out

(a) $7 + 4 \times 5$ (b) $(9 - 3)^2 \div 4$

2.2 Work out

(a) $20 - 3 \times 6$ (b) $(2 + 5)^2 - 9$

2.3 Work out

(a) $12 + 18 \div 3$ (b) $(14 - 6)^2 \div 8$

2.4 Work out

(a) $6 + 5 \times 3$ (b) $(11 - 5)^2 \div 9$

2.5 Work out

(a) $30 - 4 \times 5$ (b) $(3 + 4)^2 - 10$

2.6 Work out

(a) $8 + 24 \div 4$ (b) $(15 - 7)^2 \div 4$

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

Worked example: rounding

Round 6.3758

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

(a)	**Step 1** Next digit is 7: round up	6.4
(b)	**Step 1** Next digit is 5: round up	6.38

Practice: rounding

2.7 Round 5.2739

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

2.8 Round 18.4615

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

2.9 Round 9.0847

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

2.10 Round 3.6582

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

2.11 Round 27.354

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

2.12 Round 4.9963

(a) correct to 1 decimal place. (b) correct to 3 significant figures.